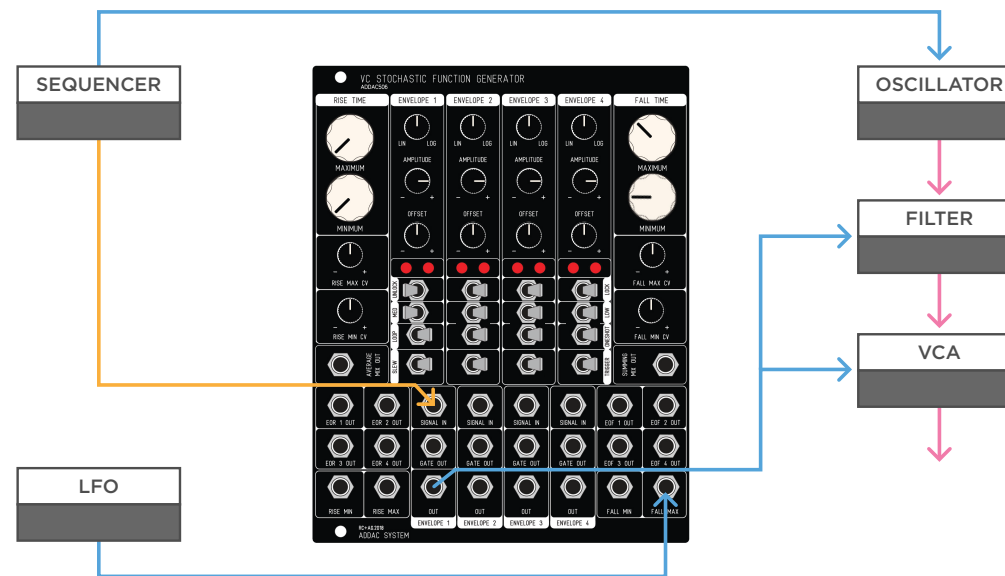


ADDAC506 Stochastic Function Generator

Stochastic Patches

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - Let's start with a simple trigger to the input in trigger mode, creating a simple attack decay envelope, going to a VCA and filter in a melodic voice. With the max fall time at zero, you can set the decay manually with the minimum fall time. When increasing the max above the minimum, you introduce some random variation to the decay of the envelope.

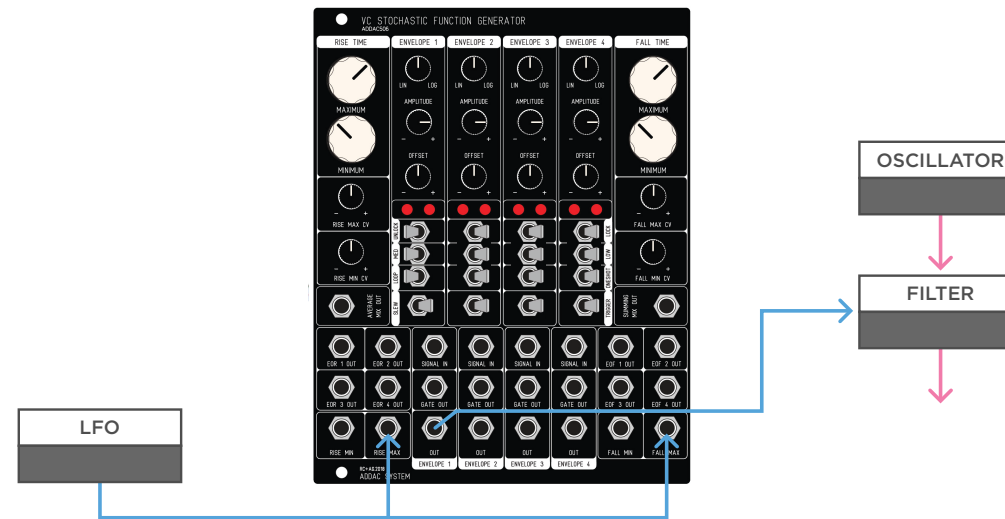
With something like a sine LFO to the minimum fall time, you can control the decay time with the LFO. But when you modulate the max fall time in this setup, you control the range in which randomization of the decay time is allowed.

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Stochastic Patches

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(a)

a - Similar tricks can be done in loop mode, creating an LFO. For example, set equal minimal rise and fall times to create a triangle LFO, influencing a filter in a drone voice. When you increase both the maximum times, you allow randomization in shape and frequency.

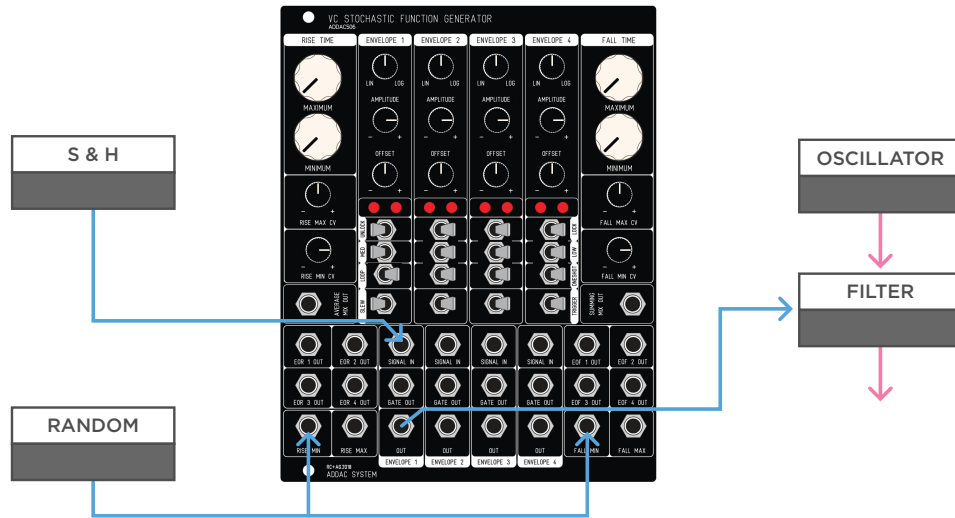
When you multiply a bipolar triangle LFO and send a copy to each of the minimum rise and fall times. Using the bipolar attenuators on each input, you can decide if you like to control the frequency of the LFO, or the shape. Again, using that same LFO on the maximum values, you can create changes in the amount of randomization allowed in generating the LFO.

ADDAC506 Stochastic Function Generator

Stochastic Patches

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MONOTRAIL TECH TALK



(a)

a - These tricks can be used on bipolar voltages, like a sample & hold signal as well. For example, send a random voltage into the minimum value inputs, to randomly morph between the stepped incoming signal, and a slewed more smooth random voltage.



(b)

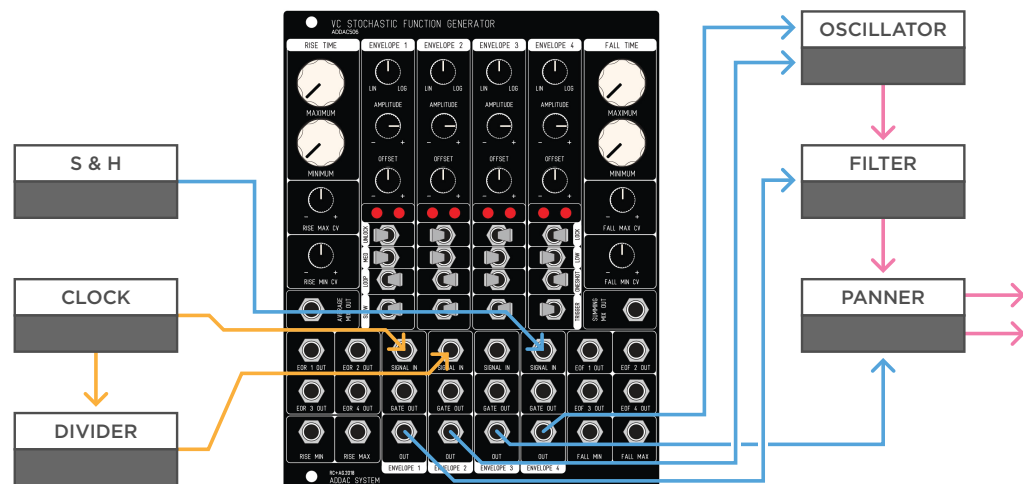
b - If you feed a slow square wave LFO into the input and set both rise and fall parameters so a bit of randomization occurs, you create some random slew, creating waveshaping on the LFO.

ADDAC506 Stochastic Function Generator

Multi Channel Patches

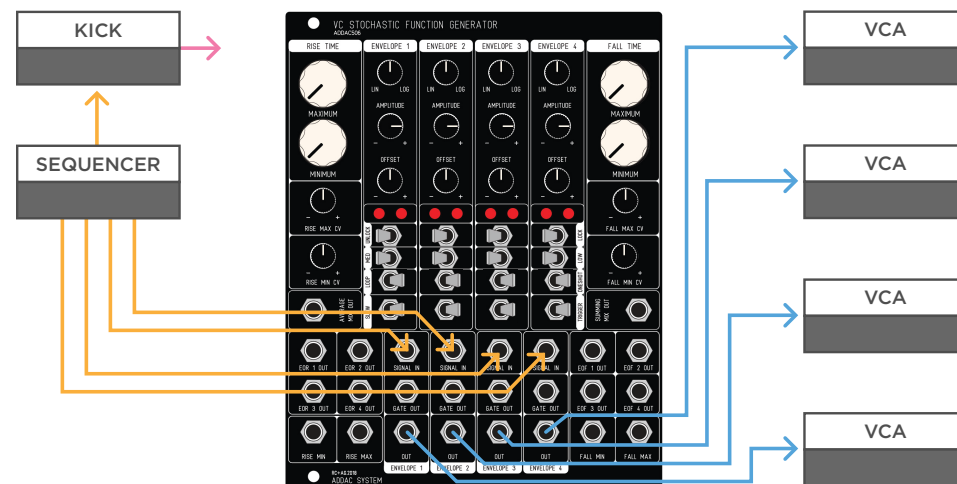
MONOTRAIL TECH TALK

Audio
Control Voltage
Trigger / Gate



(a)

a - In this example the four channels are used in a single rhythmic voice. One channel is locked, fed a sample & hold signal, and used as a slow limiter, folding the waveshape of an oscillator. Another channel is looping, also locked, and panning the signal left and right. Then a clock is driving a clock divider. The faster division is triggering an envelope, opening the filter. And a slower division is changing another folding parameter on the oscillator. Those two envelopes have a bit of randomization going on, to create organic variations.



(b)

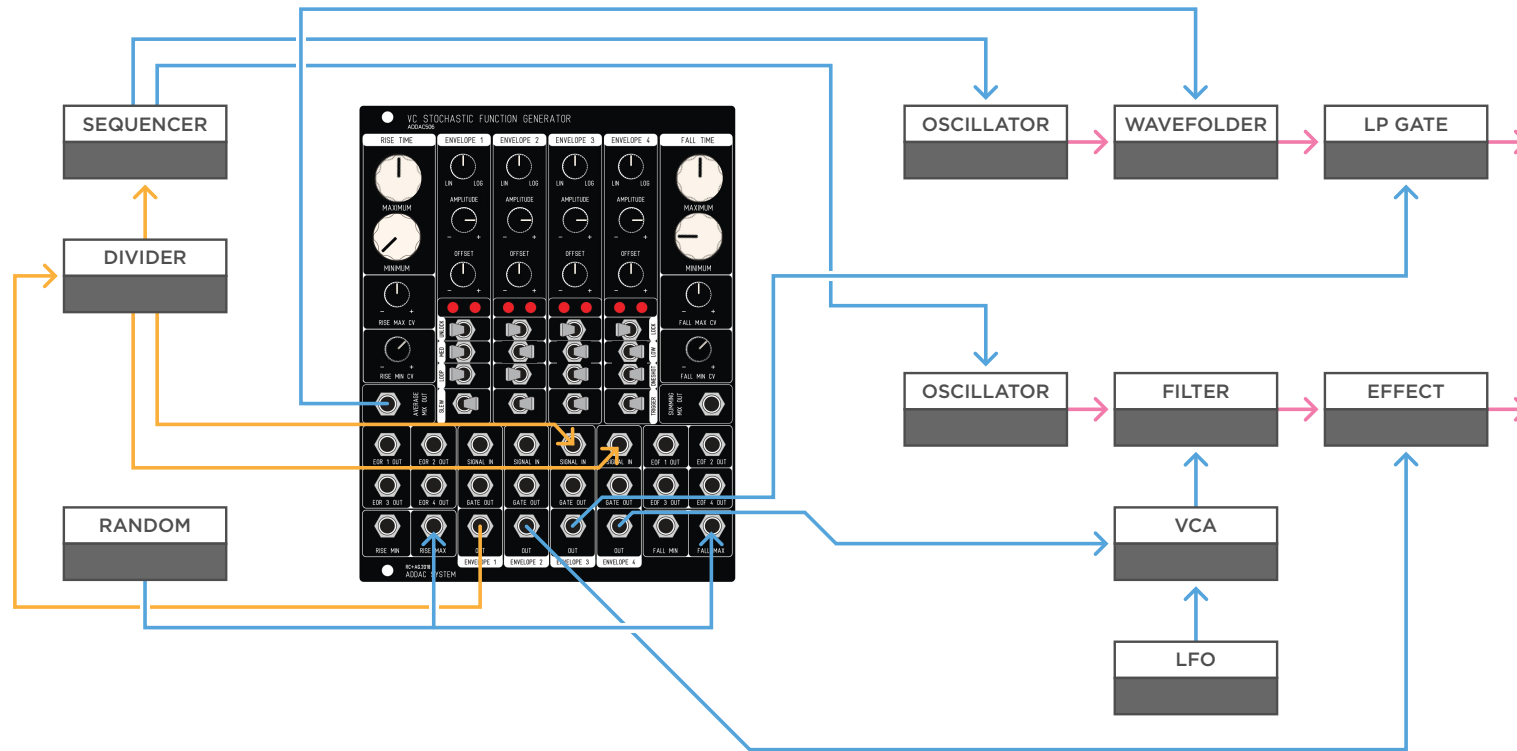
b - A nice percussive setup is to use a sequencer to create multiple trigger patterns. One is triggering a four to the floor kick, and four other patterns are used to trigger envelopes on the ADDAC506. Those envelopes are send to VCAs, shaping the volume of other percussive sounds passing through them. Like noise through a high pass filter, a tuned wavetable oscillator, and a bass sound for example.

ADDAC506 Stochastic Function Generator

Multi Channel Patches

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



(a)

a - This is a more complex ambient setup. One channel is used as a clock, driving a clock divider. One division is driving a dual 8 step sequencer, tuning two oscillators in simple voices. Two other division trigger envelopes on the ADDAC. One envelope is opening a short plucky sound on a gate, and the other opens a VCA, allowing a fast modulation LFO to pass on to a filter in the second voice.

The fourth channel is modulating an effect module. Then the average mix output is used a complex modulation source on the waveshape of an oscillator. And finally, a smooth random voltage is send to both the max rise and fall time, to allow randomization to all four channels, including the one used as the master clock.

MONOTRAIL TECH TALK

- Audio
- Control Voltage
- Trigger / Gate



a - When you use the slew function on faster signals, like audio signals, or fast LFO's, increasing the rise and fall time act as rudimentary filter or VCA. For example, if you send a sample into the input, you can set the rise and fall time so slow, the incoming signal is slewed so much, it's no longer audible. Now, when you send an envelope to both the rise and fall minimum, and use the attenuverters to invert the effect of the envelope. It will effectively reduce the slew amount, until the incoming audio is passed on, creating a VCA effect.



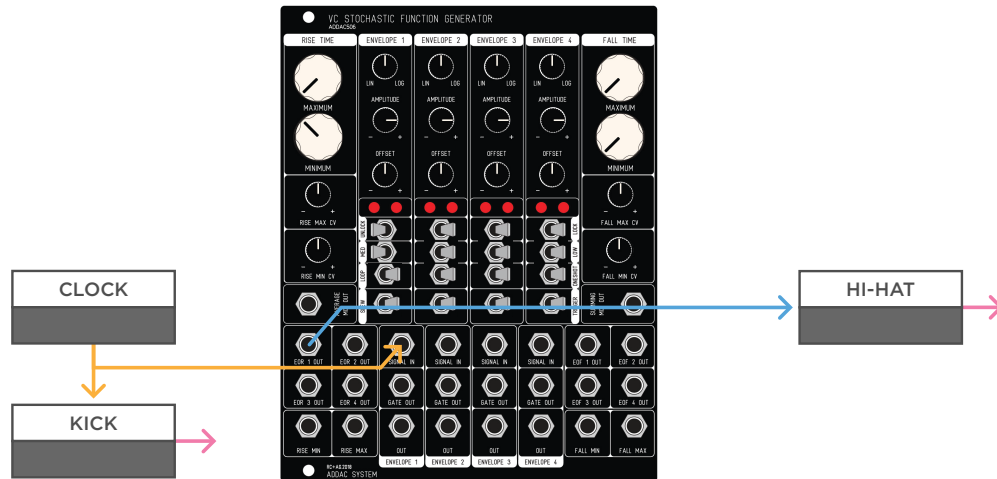
b - The extra gate and end of phase outputs allow for creative trigger and gate generation. In loop mode the gate output can be used as a square wave LFO with variable duty cycle, or when randomization is allowed, random gates.

ADDAC506 Stochastic Function Generator

Bonus Ideas

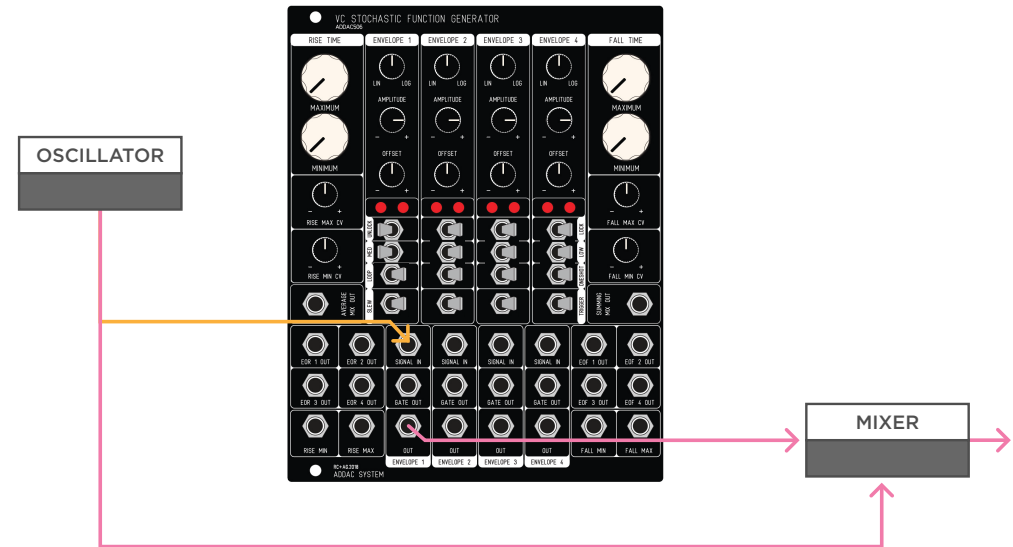
- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL TECH TALK



(a)

a - When you feed a steady clock into the input in trigger mode, the end of rise trigger output can be used as a trigger delay. For example when you use a steady clock to trigger a kick, you can use the trigger delay to trigger a high-hat, and set a manual swing. In this setup you can use the max rise value to create some subtle randomization if you like more organic wobbly beats. When you set the rise time above the speed of the incoming clock, you can set up a manual clock divider. For example, create a divide by two, or three.



(b)

b - if you trigger a channel with an oscillator, a function generator can also be used as audio source, or experimental sub oscillator with wonky waveshaping effects. Let's just scratch the surface of that by using a mixer to mix the main oscillator with the output of the function generator, and tweak some knobs by hand